

CURIONEST

CHEMISTRY MEASUREMENT

Lab Tools, Density & Percent Error

Visual worksheet for U.S. Grades 8-10

STUDENTS WILL

- identify common measurement tools
- justify equipment choices with visual evidence
- calculate density and percent error
- compare accuracy and precision
- repair a measurement misconception



4 student pages
4-page full key
Editable DOCX

PRINT-READY • US LETTER

$$\text{density} = \text{mass} / \text{volume}$$

TEACHER QUICK START

Chemistry Measurement Lab Tools & Density Visual Worksheet

AT A GLANCE

U.S. Grades 8-10 | 35-45 minutes | Individual, pairs, or intervention | No lab materials required

Purpose	Students connect visible equipment features to measurement choices, then apply the resulting data to density, percent error, accuracy, and precision.
Suggested pacing	5 min quick reference; 10 min tool selection; 15 min calculations; 10 min lab case and repair; 5 min review.
Before class	Print pages 1-4. Decide whether students may use calculators. Review local expectations for significant figures and liquid-volume reading.
Differentiation	Support: allow the formula toolbox and a unit bank. Core: complete as written. Extension: ask students to design one additional measurement trial that would improve confidence in precision.
Look-fors	Students name the measured quantity, not only the tool; carry units through calculations; use the accepted value in the percent-error denominator; avoid claiming precision from a single trial.
Safety note	This is a paper-based readiness task. Local laboratory rules, teacher directions, SDS information, and equipment-specific procedures always take priority.

Fast formative check

Ask: "What makes a measurement trustworthy?" A complete response should connect the right tool, readable scale, unit, repeated trials, and transparent calculation.